Exploring the pangenome of *Candidatus* Atelocyanobacterium thalassa (UCYN-A): Implications for endosymbiosis and evolution as a nitrogen-fixing organelle (nitroplast)

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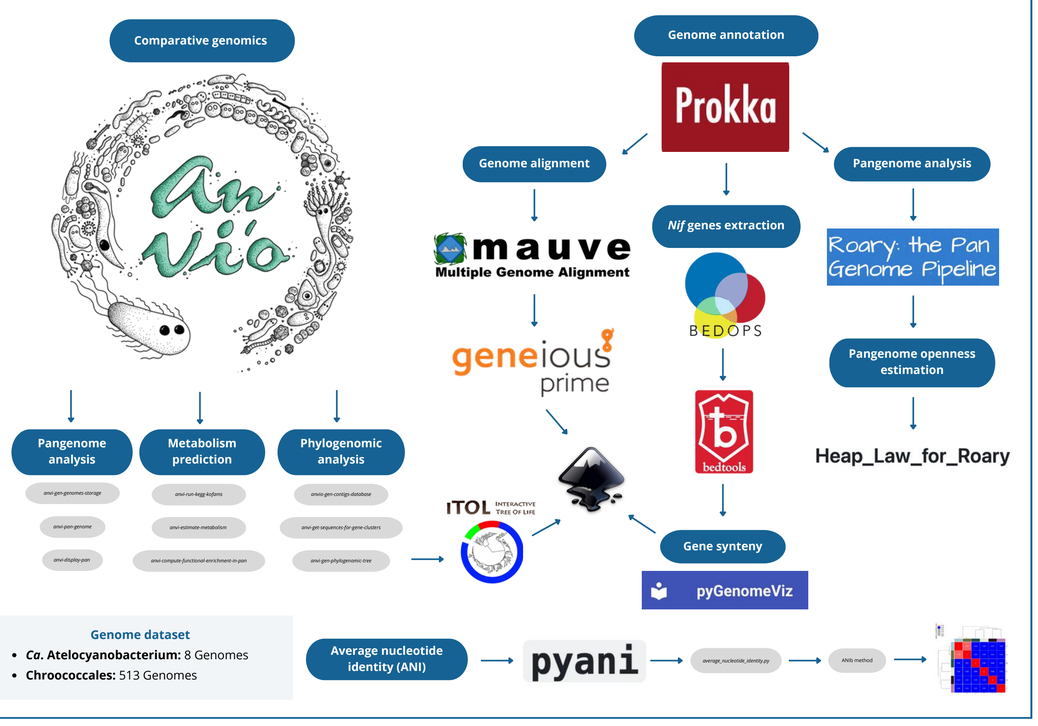
Introduction

The diazotroph *Candidatus* Atelocyanobacterium thalassa (UCYN-A) is an endosymbiont of the marine alga *Braarudosphaera bigelowii*, currently in an evolutionary transition towards becoming an organelle specialized in nitrogen fixation, known as a "nitroplast" (Massama Ramon, 2024). Previous estimates of nitrogen fixation in the oceans focused on filamentous cyanobacteria such as *Trichodesmium* and *Richelia* (Massama Ramon, 2024; Vieyra-Mexicano et al., 2024). However, the discovery of UCYN-A has revealed that this diazotroph is a key component in global N2 fixation (Bombar et al., 2014). In this study, we present the first pangenome constructed for UCYN-A, providing significant insights into the genomic adaptations that enable it to maintain its close relationship with its host.

Materials and methods



Candidatus Atelocyanobacterium have a closed pangenome

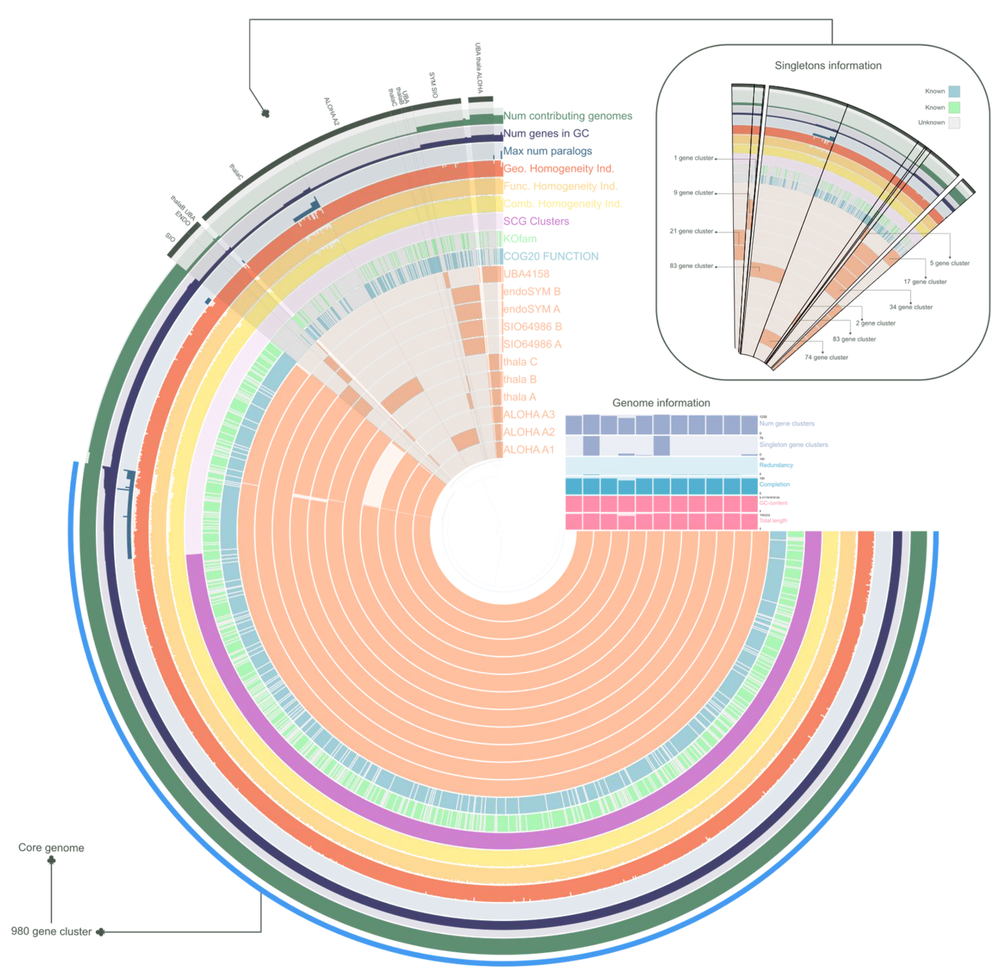


Figure 1. Overview of the *Ca.* Atelocyanobacterium pangenome structure. This figure illustrates the genomes organized according to a gene presence/absence clustering tree. The outer rings show details of single-copy gene clusters, including gene counts, the genomes they contribute to, and homogeneity indices. Bar charts provide key genomic metrics. In the upper right section, gene clusters that are specific to certain genomes are highlighted. The estimated Heaps' law exponent (λ) was 0.3.

Conclusions

This study provides insight into the genomic and metabolic adaptations of UCYN-A. The pangenomic and phylogenomic analyses reveal previously unrecognized diversity, and the data support the idea of a metabolic dependency between UCYN-A and its algal host (*B. bigelowii*), specifically in metabolic pathways related to carbon, amino acids, and other essential molecules. These findings not only help us better understand the role of UCYN-A in marine ecosystems but also open new avenues for future research on the evolution of complex symbioses and their applications in biotechnology and sustainable agriculture.

Two distinct phylogroups of *Ca.* Atelocyanobacterium: Proposal of a novel species

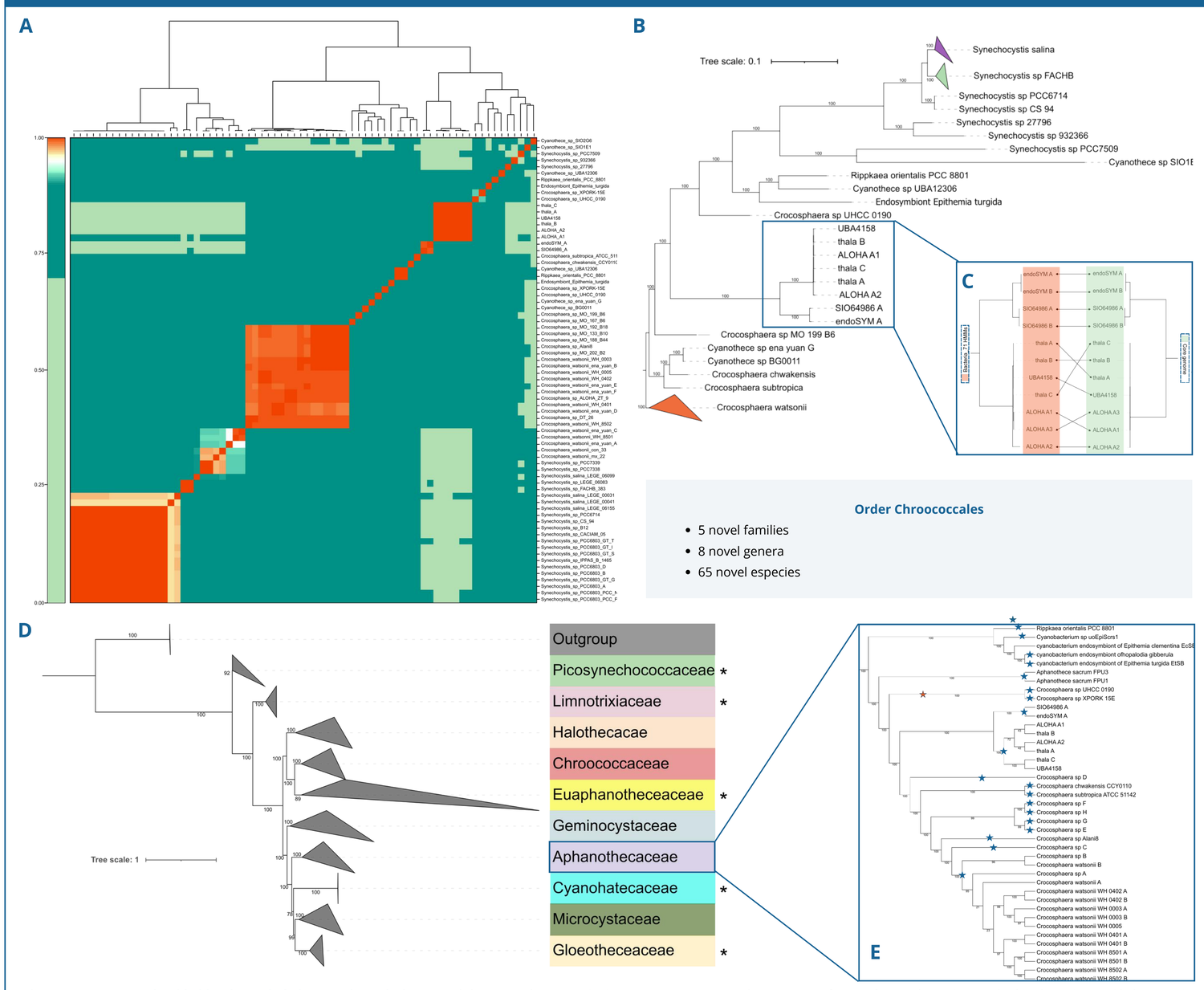


Figure 2. Average nucleotide identity and phylogenomic analysis. (A) ANIb heatmap showing pairwise comparisons among the genomes of *Ca.* Atelocyanobacterium and related species **(B)** Phylogenomic tree constructed using 71 single-copy core genes (Bacteria_71). **(C)** Tanglegram comparing phylogenomic tree based on the core genome and the tree constructed using the Bacteria_71 dataset. **(D)** Phylogenomic tree of the order Chroococcales based on Bacteria_71, including 513 genomes. Newly proposed families are indicated by an asterisk (*). **E)** Expanded view of the family *Aphanothecaceae*, showing the two species of *Ca.* Atelocyanobacterium alongside 13 novel species and one newly proposed genus. The clade containing the novel species are marked with a blue star ( ), while the novel genus is indicated by an orange star ( ).

Complete nitrogenase enzyme complex: *nif* operon

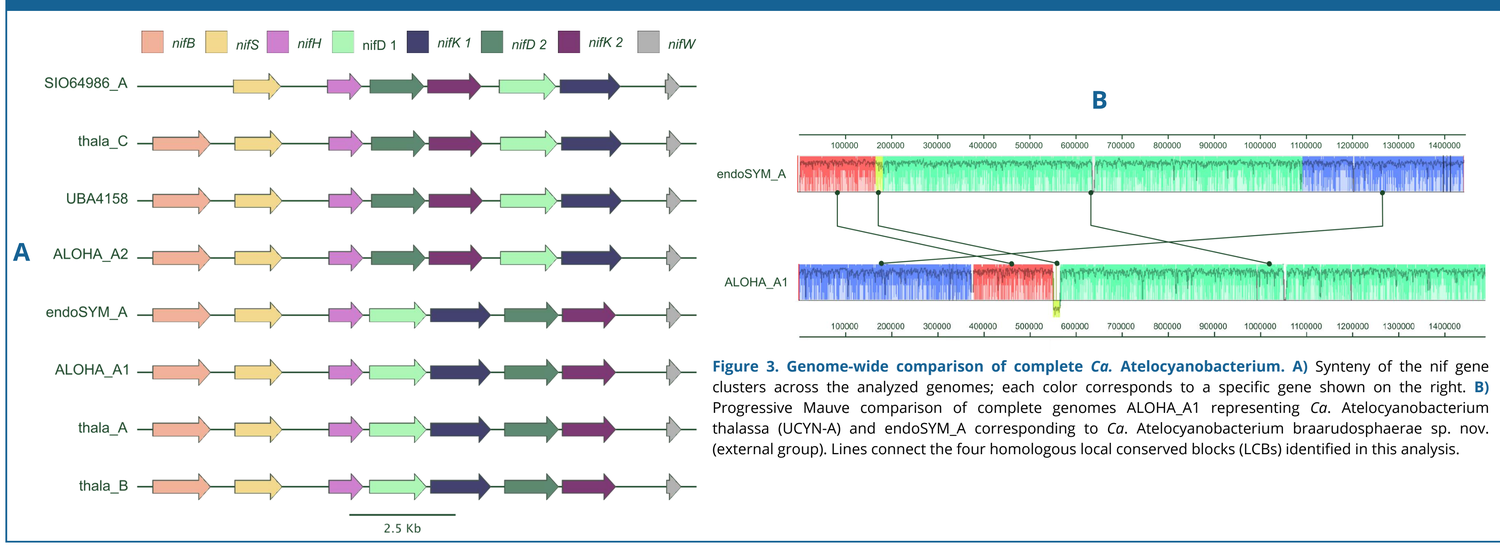
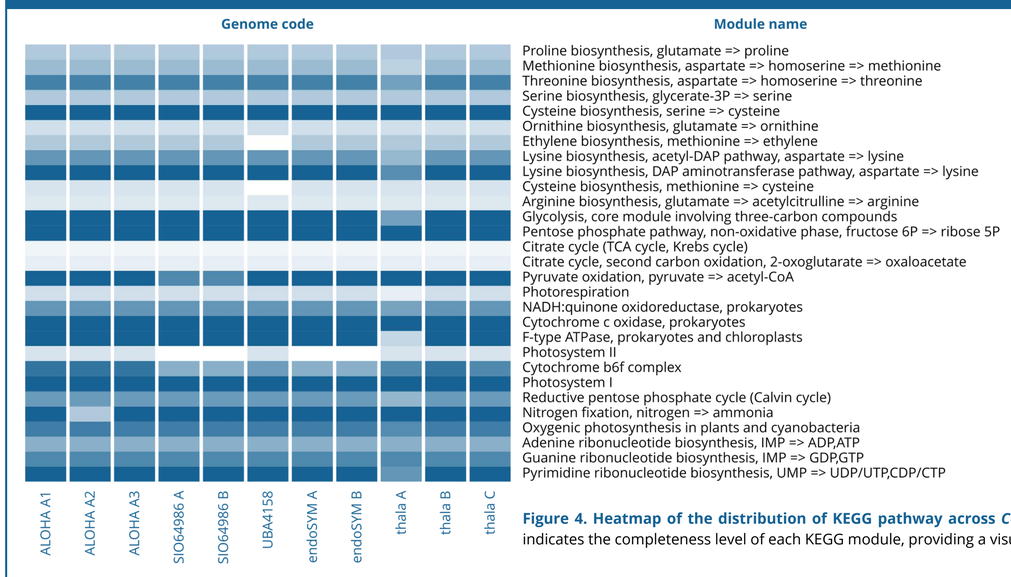


Figure 3. Genome-wide comparison of complete *Ca.* Atelocyanobacterium. A) Synteny of the nif gene clusters across the analyzed genomes; each color corresponds to a specific gene shown on the right. **B)** Progressive Mauve comparison of complete genomes ALOHA_A1 representing *Ca.* Atelocyanobacterium thalassa (UCYN-A) and endoSYM_A corresponding to *Ca.* Atelocyanobacterium braarudosphaerae sp. nov. (external group). Lines connect the four homologous local conserved blocks (LCBs) identified in this analysis.

Metabolism prediction reveals insights into the metabolic dependency between *Ca.* Atelocyanobacterium and *B. bigelowii*



Complete pathways

- Nitrogen fixation
- Cytochrome c oxidase, F-type ATPase, NADH oxidoreductase, and NAD(P)H oxidoreductase
- Cysteine biosynthesis from serine to cysteine
- Guanine ribonucleotide and adenine ribonucleotide biosynthesis

Incomplete pathways

- Oxygenic photosynthesis
- Reductive pentose phosphate cycle (Calvin cycle)
- Photorespiration
- Proline, threonine, arginine, cysteine, ethylene and methionine biosynthesis
- Citric acid cycle (Krebs cycle)

Figure 4. Heatmap of the distribution of KEGG pathway across *Ca.* Atelocyanobacterium genomes. The intensity of the blue color indicates the completeness level of each KEGG module, providing a visual assessment of pathway integrity within each genome.

References

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2. Vieyra-Mexicano C, Souza V, Pajares S. Distribution of the N2 -fixing cyanobacterium *Candidatus* Atelocyanobacterium thalassa in the Mexican Pacific upwelling system under two contrasting El Niño Southern Oscillation conditions. Environ Microbiol Rep. 2024 Feb 13;16(1).
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